Algorithms as Recipes

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Algorithms as Recipes before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Features: Useful Clues

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Sound Becomes a Signal

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sensors Are Messengers

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Algorithms as Recipes trace and record the first step you can explain without help.

Use W2: Distinguish replacement from addition for **Algorithm**s and changing **state**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Algorithms as Recipes explanation against the supplied facts before discussing it.

Use the worked case for **Algorithm**s and changing **state**. Proposed explanation: “Every instruction uses the original number”. Decide whether this explanation is supported.

Point to the **step** or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Trace two changing number boxes

For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction. Explain why Y does not become 7 when X changes.

Then swap the first two instructions and **trace** the new procedure from the same **initial state**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Algorithms as Recipes target to a trace and a justified conclusion.

On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. **Trace** position and direction after every instruction. What if TURN were unspecified?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Algorithms as Recipes case and show the deciding evidence.

A robot starts on the left of three squares, facing right. It executes MOVE, MOVE, STOP, each MOVE being one square.

Trace the position after each command.

Show your **trace**, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Instructions say pick up a pencil, write your name, then find paper. Find the first ordering problem and rewrite the three **step**s so another learner can follow them.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Algorithms as Recipes adds to Helpful Classroom Sorter and keep the evidence.

Write the first executable sorting **algorithm** for one easy item before adding branches or AI.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next **step** _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. **Trace** position and direction after every instruction. What if TURN were unspecified?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Trace the current value

Trace the current value. Apply the supplied Algorithms as Recipes facts and keep your reasoning.

Start score at 3. Add 4 to score, double score, then subtract 5. Record the **initial state** and the **state** after each instruction. Underline the number used by the doubling **step** and explain where it came from.

My reasoning and evidence

W2 Distinguish replacement from addition

> Distinguish replacement from addition. Apply the supplied Algorithms as Recipes facts and keep your reasoning.

Two procedures start score at 6. Procedure A sets score to 4, then adds 2. Procedure B adds 4 to score, then adds 2. **Trace** both and explain why the first instructions are different.

My reasoning and evidence

W3 Repair the first error

> Repair the first error. Apply the supplied Algorithms as Recipes facts and keep your reasoning.

A procedure starts score at 6, adds 2, doubles score, then subtracts 3. A **trace** says 6, 8, 12, 9. Identify the first wrong transition, explain the likely mistake and write a complete corrected **trace**.

Why is changing only the last value inadequate?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Trace two changing number boxes

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Trace two changing number boxes. Apply the supplied Algorithms as Recipes facts and keep your reasoning.

For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction. Explain why Y does not become 7 when X changes.

Then swap the first two instructions and **trace** the new procedure from the same **initial state**.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Test what a final answer can prove

> Test what a final answer can prove. Apply the supplied Algorithms as Recipes facts and keep

> your reasoning.

The required procedure starts score at 10, adds 2 and subtracts 2. A learner instead adds 4 and subtracts 4. Both report 10.

Write both **trace**s and explain whether matching the final result proves that the learner followed the required procedure.

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

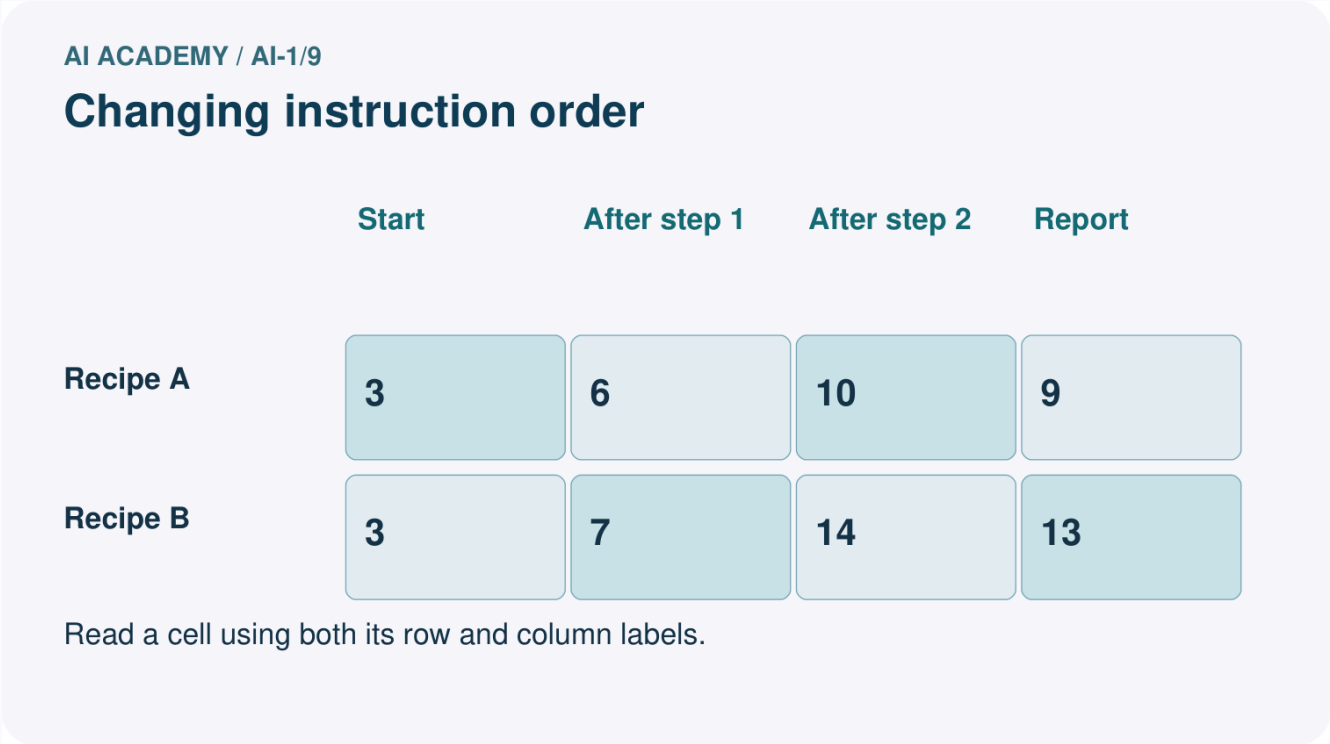
Pictorial case practice: Algorithms as Recipes

> A fictional paper badge machine transforms an initial score using an ordered recipe. Explain the labelled step before checking the changed condition.

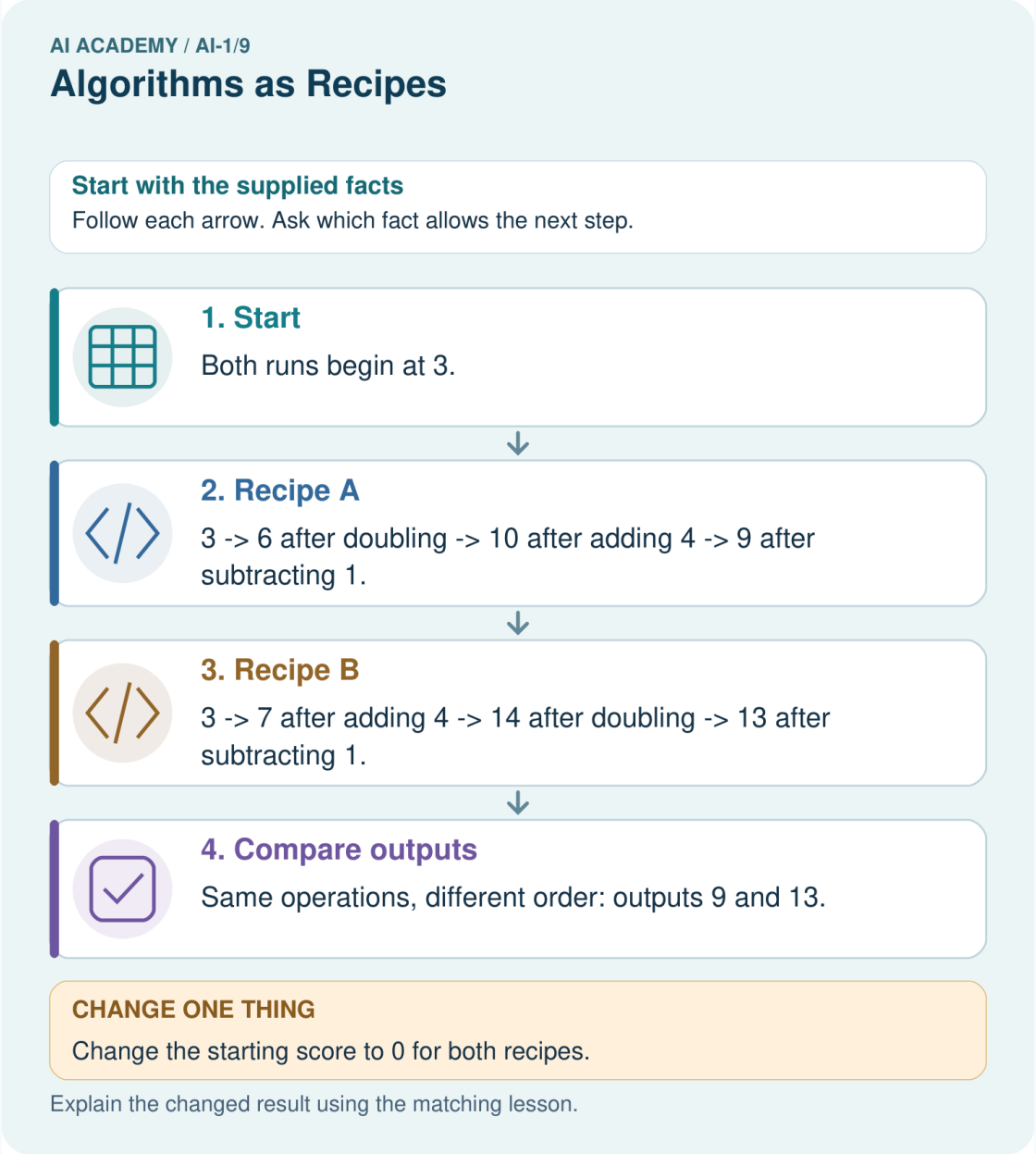
A fictional paper badge machine transforms an initial score using an ordered recipe.

Start score=3. Recipe A: double, add 4, subtract 1, report.

Recipe B swaps the first two operations but keeps the same starting score.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Use the arrows to explain where the two zero-start runs first differ.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Write the sorter's **step**s in order, showing which values each **step** updates so a partner can reproduce the result.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a **trace** of how one **input** becomes an **output**; explain the rule or learned component.

Write the sorter's **step**s in order, showing which values each **step** updates so a partner can reproduce the result.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper badge machine transforms an initial score using an ordered recipe.

Practice 1: explain the deciding evidence

Which labelled **step** matches this detail? Same operations, different order: **output**s 9 and 13.

1. Recipe A

2. Compare **output**s

3. Recipe B

4. Start

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the **step**s in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare **output**s Same operations, different order: **output**s 9 and 13. Every instruction reads the current score, not the original one unless explicitly told to.

In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling. The final difference of 4 follows from that order, not from different starting data.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change the starting score to 0 for both recipes. Which conclusion is supported by the supplied facts?

1. A reports 3 and B reports 7.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding **step** in the pictorial **trace**. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A reports 3 and B reports 7. A: 0→0→4→3. B: 0→4→8→7.

Repair a missing building block

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a **state**d task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered **sequence** of sound-signal samples with a **state**d time interval. Short practice: Does the all-zero retained record prove the full signal was silent?

Use the picture.. Return to this week and explain what you changed.

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

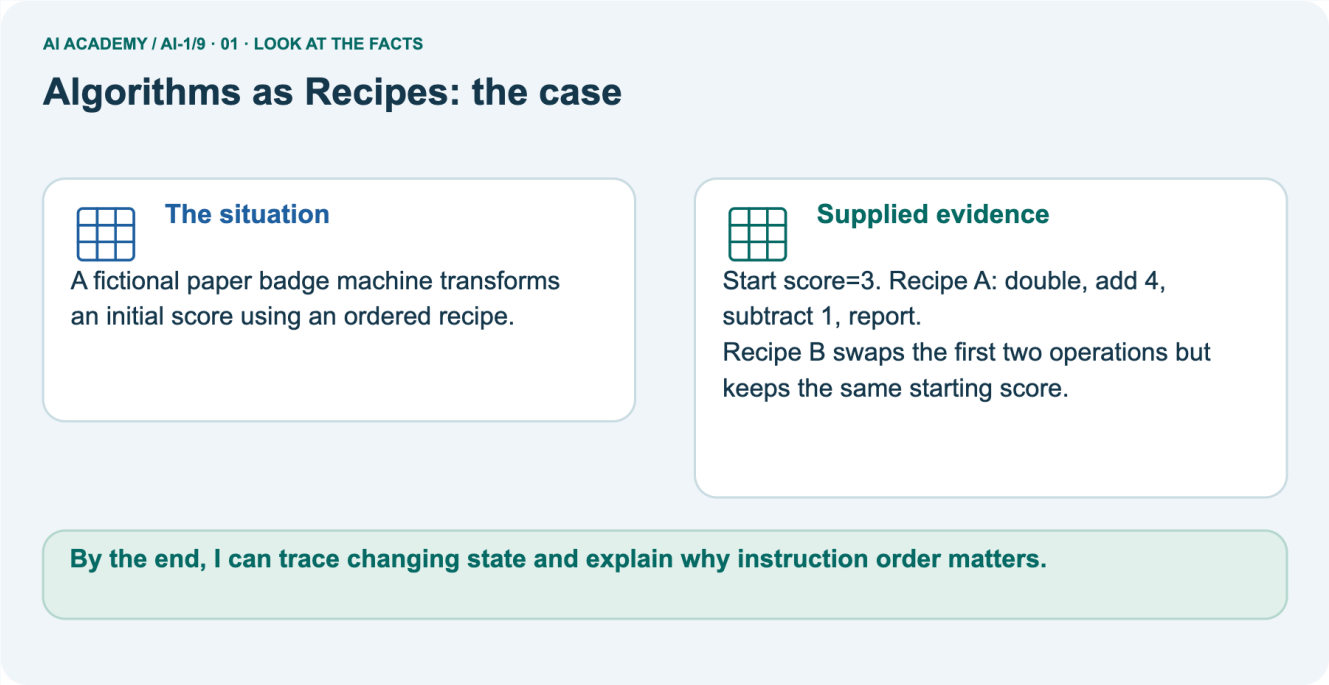
> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Algorithms as Recipes picture from its labels, then record a relevant limit.

01. Algorithms as Recipes: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

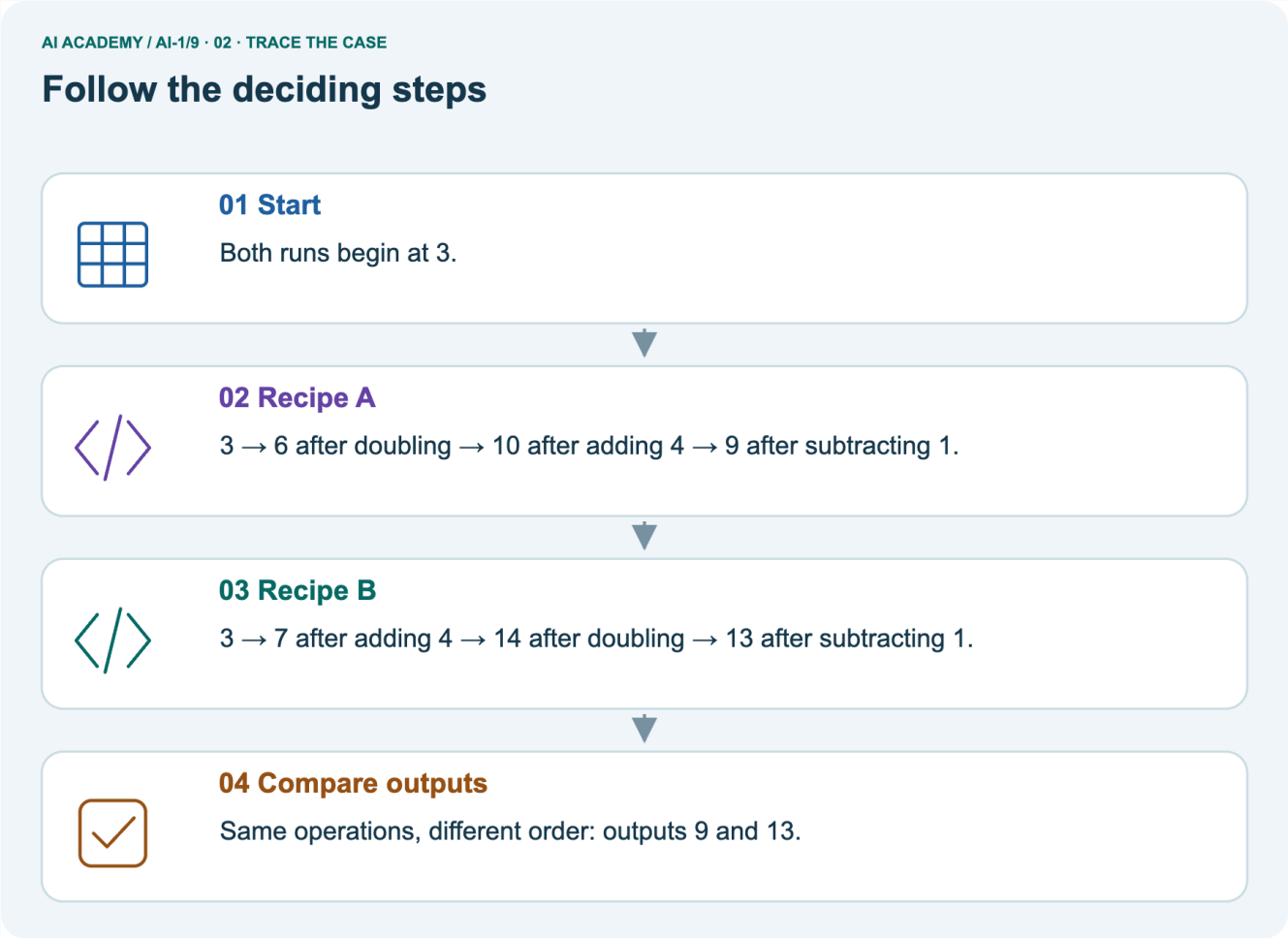
A fictional paper badge machine transforms an initial score using an ordered recipe. Supplied facts: Start score=3. Recipe A: double, add 4, subtract 1, report.

Recipe B swaps the first two operations but keeps the same starting score.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



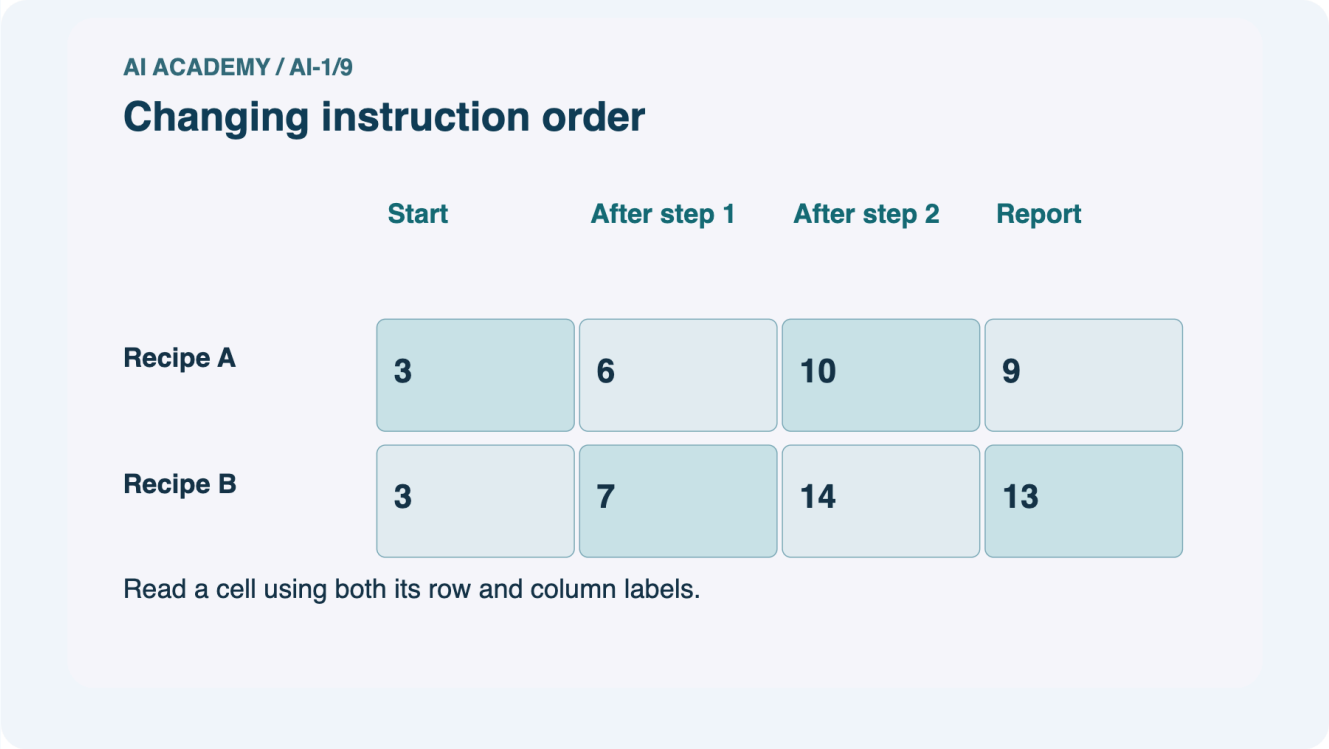
Follow the labelled picture with the matching explanation. Open the vector diagram

Start: Both runs begin at 3. → Recipe A: 3 → 6 after doubling → 10 after adding 4 → 9 after subtracting 1. → Recipe B: 3 → 7 after adding 4 → 14 after doubling → 13 after subtracting 1. → Compare **output**s: Same operations, different order: **output**s 9 and 13.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Changing instruction order



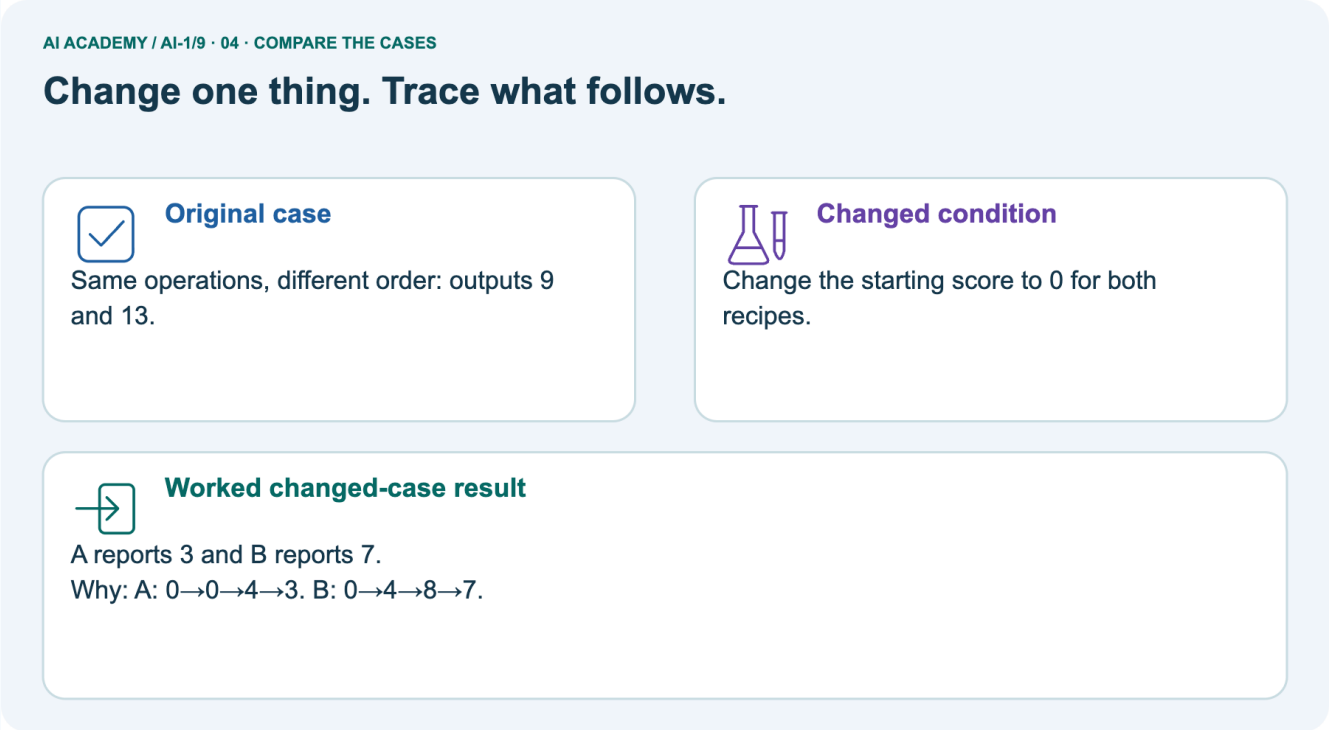
Follow the labelled picture with the matching explanation. Open the vector diagram

Changing instruction order. Recipe A: Start = 3, After **step** 1 = 6, After **step** 2 = 10, Report = 9. Recipe B: Start = 3, After **step** 1 = 7, After **step** 2 = 14, Report = 13.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



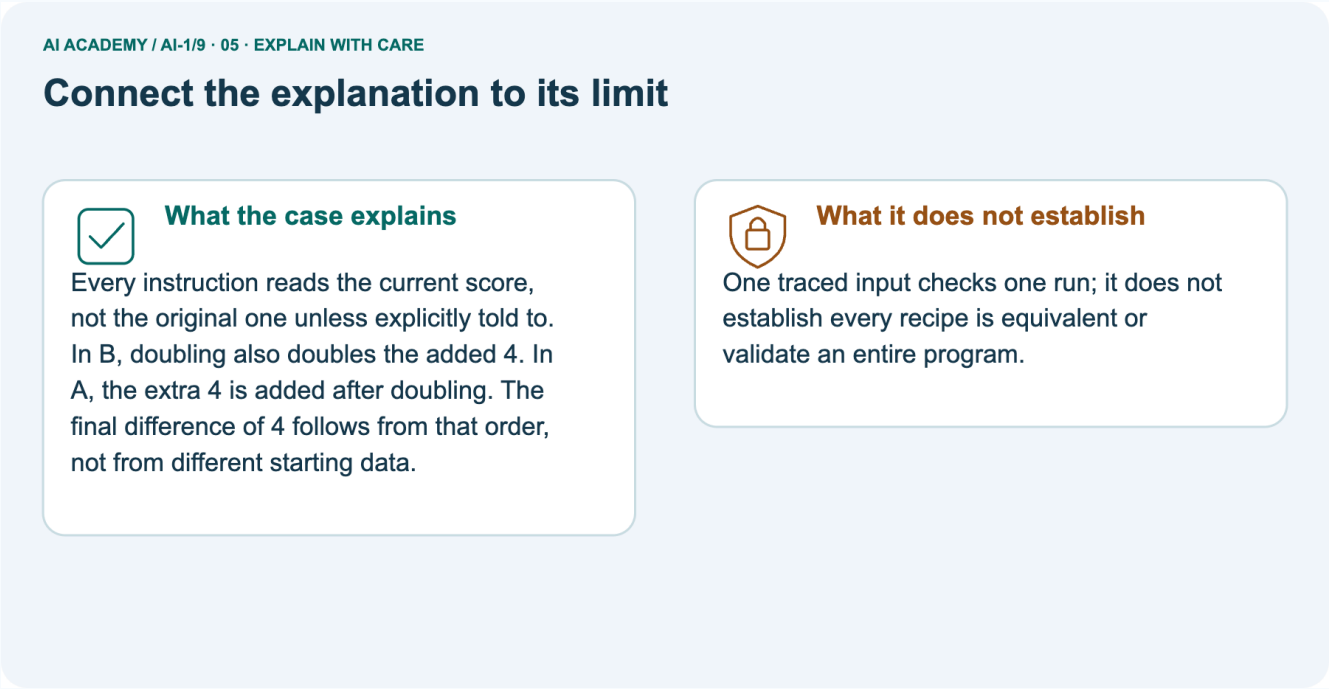
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Same operations, different order: **output**s 9 and 13. Changed condition: Change the starting score to 0 for both recipes. Expected result: A reports 3 and B reports 7. Reason: A: 0→0→4→3. B: 0→4→8→7.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

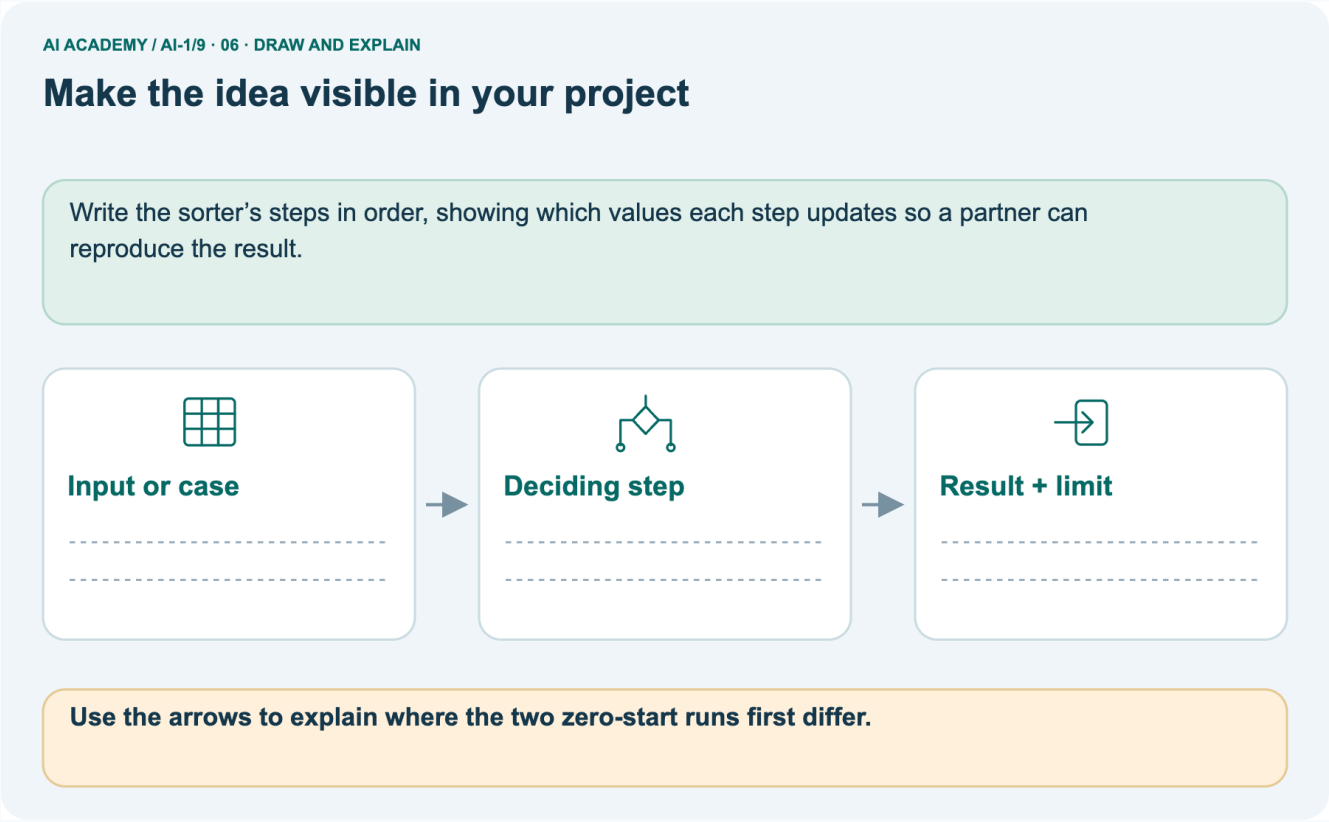
Every instruction reads the current score, not the original one unless explicitly told to. In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling.

The final difference of 4 follows from that order, not from different starting data. Limit: One **trace**d **input** checks one run; it does not establish every recipe is equivalent or validate an entire program.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write the sorter's **step**s in order, showing which values each **step** updates so a partner can reproduce the result. Draw your **input**, deciding **step**, result and limit.

Supported question: Use the arrows to explain where the two zero-start runs first differ.

Draw your project **trace** and justify the deciding **step** using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.